

Single Cell RNA Sequencing Workflow for 10X Human Glioblastoma Multiforme Dataset

ABSTRACT

The aim of this project is to describe a methodology for single cell RNA sequencing within RStudio 4.5.0 using addons such as Seurat, DoubletFinder and SingleR to create a flexible process that can be repurposed for a variety of cell types. Data is outputted in an easily readable, but simultaneously editable format in order to minimise unnecessary repetition of code for different cell types. To achieve this, several existing methodologies and libraries have been utilised, with each referenced below.

The resulting script, methodology and figures are available on the repository as an example and are intended to be modified accordingly to fit the specific cell type and purpose of a study or project.

INTRODUCTION

RNA Sequencing of cells allows measuring of gene expression activity within various cell samples. ScRNASeq allows comparison between the transcriptomes of individual cells, allowing for greater resolution than higher-level techniques such as Bulk RNA Seq. Many methods exist when undertaking ScRNASeq, however all begin by generating the raw sequence data. This is always stored in a feature-barcode sparse matrix, and represented by the common file formats .h5, .hdf5, .rds, .h5ad, .loom and .mtx. After the raw data is obtained in a relevant file format, the described methodology can be used to analyse it within RStudio.

The raw data used for this pipeline was obtained from the 10X genomics public library. For this dataset, cells were sequenced on the Illumina NovaSeq 6000 to a depth of 44,736 mean reads per cell. 10X Genomics also provides their own reference analysis of the dataset for comparison, available on their website. This analysis cites a cell count of 5604, with 3094 mean genes per cell and 9434 median UMIs per cell. The raw matrix provided was in the “.h5” format.

METHODOLOGY

Data Import

As data is in the “.h5” format, the Read10X_h5 function native to Seurat is used to import the data into R. When importing different datatypes, different functions are necessary to parse them correctly. Details on these functions are available in the Seurat documentation.

The imported data is then converted to a Seurat object for further manipulation and mitochondrial information is added to the metadata.

Quality Check

Code under the “Quality Check” header provides key data prior to manipulation, including a count of cell number, gene count, UMI count and mitochondrial percentage distribution.

The cell number is important as any more cells present than were loaded will be “junk” cells and compromise the accuracy of the data. This data also allows spotting of potential doublets by finding cells that are anomalously high in molecule or gene counts. The percentage of mitochondrial genes are calculated and visualised in order to tag cells that are dying, as these genes are often upregulated when this occurs.

The graph on line 32 visualises the number of genes against number of UMIs and the mitochondrial percentage. Cells should fit approximately along the $x=y$ line, with few or no cells appearing in the top left or bottom right corners.

Also included is a graph showcasing the distribution of gene complexity.

Filter and Preprocessing

Low quality cells are filtered out using their mitochondrial expression percentage, number of features and count of features. High rarity genes are also filtered out in order to improve data readability. Data is then normalised using Seurat in order to allow comparison of expression across multiple cells. The top variable features are then identified through Seurat, stored and visualised.

Prior to clustering, the data must be scaled as there will be unwanted sources of variation present due to either cell cycle phase differences or batch number differences. If this step is not performed, cells would cluster based on these factors which should be avoided.

We also perform linear dimensionality reduction, which identifies sources of heterogeneity in the dataset and outputs genes with the largest diversity in expression. This is achieved by assigning principal component scores to each gene, with the output graph showing the number of principal components per standard deviation. This is because we only want to capture components with a high deviation for downstream analysis and trim off components with low standard deviations.

Clustering

This process clusters together cells with similar expression patterns as determined by our linear dimensionality reduction and previous analyses. In order to accomplish this we use the number of principal components previously calculated in the elbow plot (in our case, this was 15) to find the relevant neighbours, and then cluster using a data-specific granularity value. To find the optimal granularity value, clustering is done several times until a result with as little overlap as possible is achieved.

Doublet Filtering

In order to preserve the integrity of the data and filter out anomalous cells an attempt to exclude doublets is made using Doublet Finder. First, the optimal pK value (number of artificial doublets) is found using the process described on the DoubletFinder GitHub repository. The optimal pK value is the highest BCmetric value as plotted on the outputted figure, which in our case is 0.21. The number of homotypic doublets is then estimated according to the number of expected doublets, which is itself determined according to the number of cells loaded and recovered. DoubletFinder is then run and the doublets are visualised. These doublets are then filtered out using the metadata stored by DoubletFinder and stored in a new variable for downstream analysis.

Differentially Expressed Gene (DEG) Analysis and Annotation

First, the FindAllMarkers function is used to identify the cell types that form each cluster and set the correct default assay for your dataset (in our case, this is RNA). The FindConservedMarkers function is then used to find conserved markers for each specific cluster. To set which cluster to examine, change the Ident.1 parameter based on previous cluster visualisation. The grouping variable is then set to the controlled or treated condition. This allows us to see fold change and differential regulation of gene expression for individual genes compared to all other clusters. This data is then plotted using a Feature Plot.

Using this data combined with other information in the study, cell types can be identified using upregulation and downregulation of specific markers. Sometimes, this may require significant research depending on the cell type, however automatic cell annotation tools can assist in this process. In this script, SingleR is used to reference CellDex for the Human Primary Cell Atlas, which then automatically annotates our clusters.

Addendum: Cell Cycle Phase Annotation

Contained in a separate script is a process for cell cycle annotation. The import and preprocessing steps are identical to those described above, however following this cells are annotated based on gene expression of various cell cycle markers as being in a specific state within the cell cycle. These are then visualised as an addition to the above analysis, deemed as relevant here as it is vital data when examining cancer and tumour samples.

RESULTS

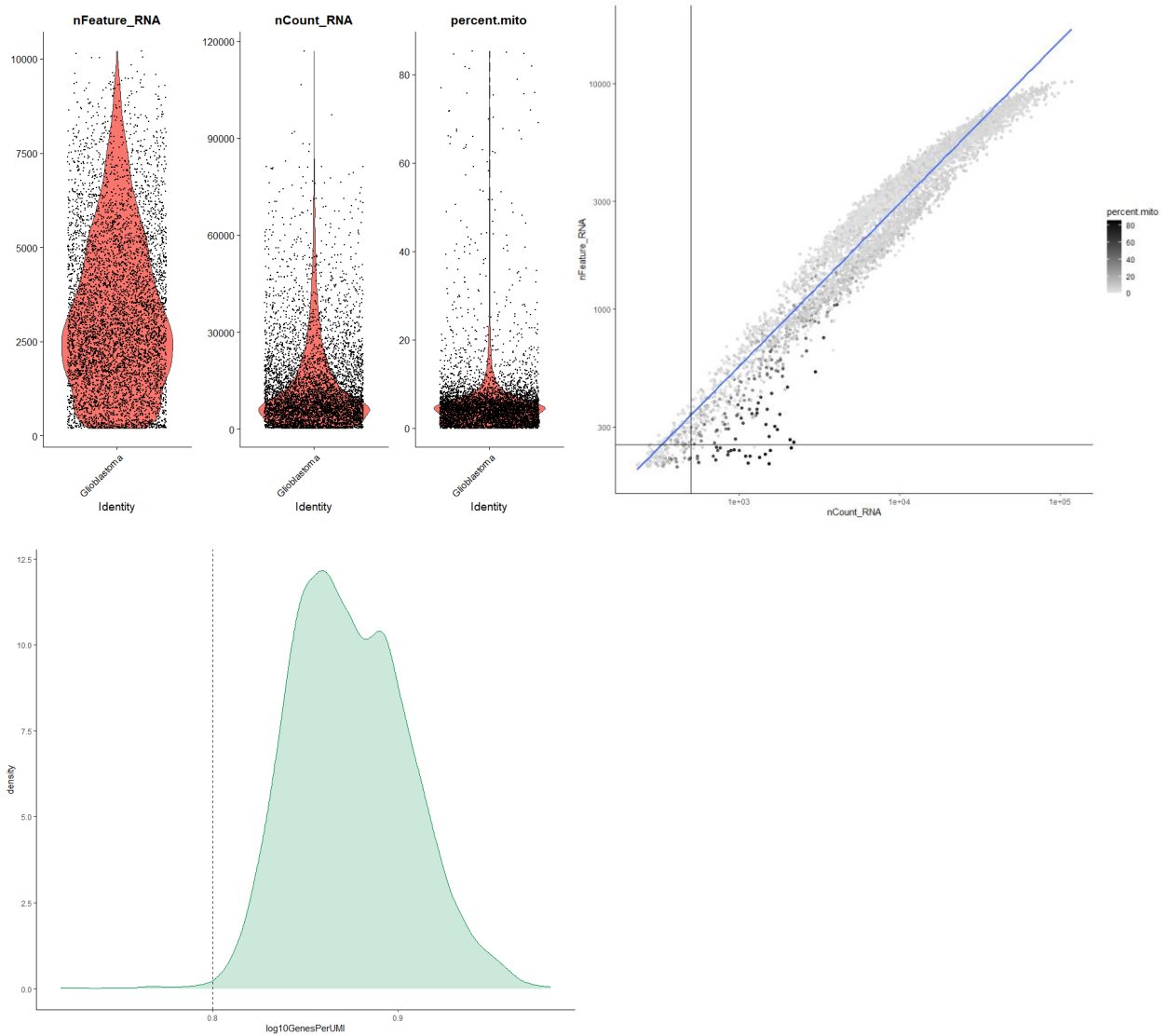


Fig. 1: A – Three violin plots showing the number of features, UMIs and percent of mitochondrial RNA respectively in the entire sample of cells. **B** – Graph showing the distribution of genes against the number of UMIs, with mitochondrial gene percentage for each cell shown by point colour as illustrated in the key. **C** – Graph showing the distribution of gene complexity across the dataset.

The results in *Figures 1A* and *1B* show key markers for interpreting the quality of the dataset. In *Figure 1A*, we see few cells with a high feature count, and the majority of cells show an expected UMI count. This indicates relatively good quality data with few doublets (although this is accounted for in greater detail using DoubletFinder). Some cells, albeit a minority, show a high percentage of mitochondrial RNA. These cells are excluded from the downstream analysis. We can see that this exclusion is a correct inference via *Figure 1B*, where the majority of cells with a high mitochondrial percentage fail to lie on the expected $x=y$ line. *Figure 1C* is also used as an approximation of dataset quality, where a rough normal distribution would be expected.

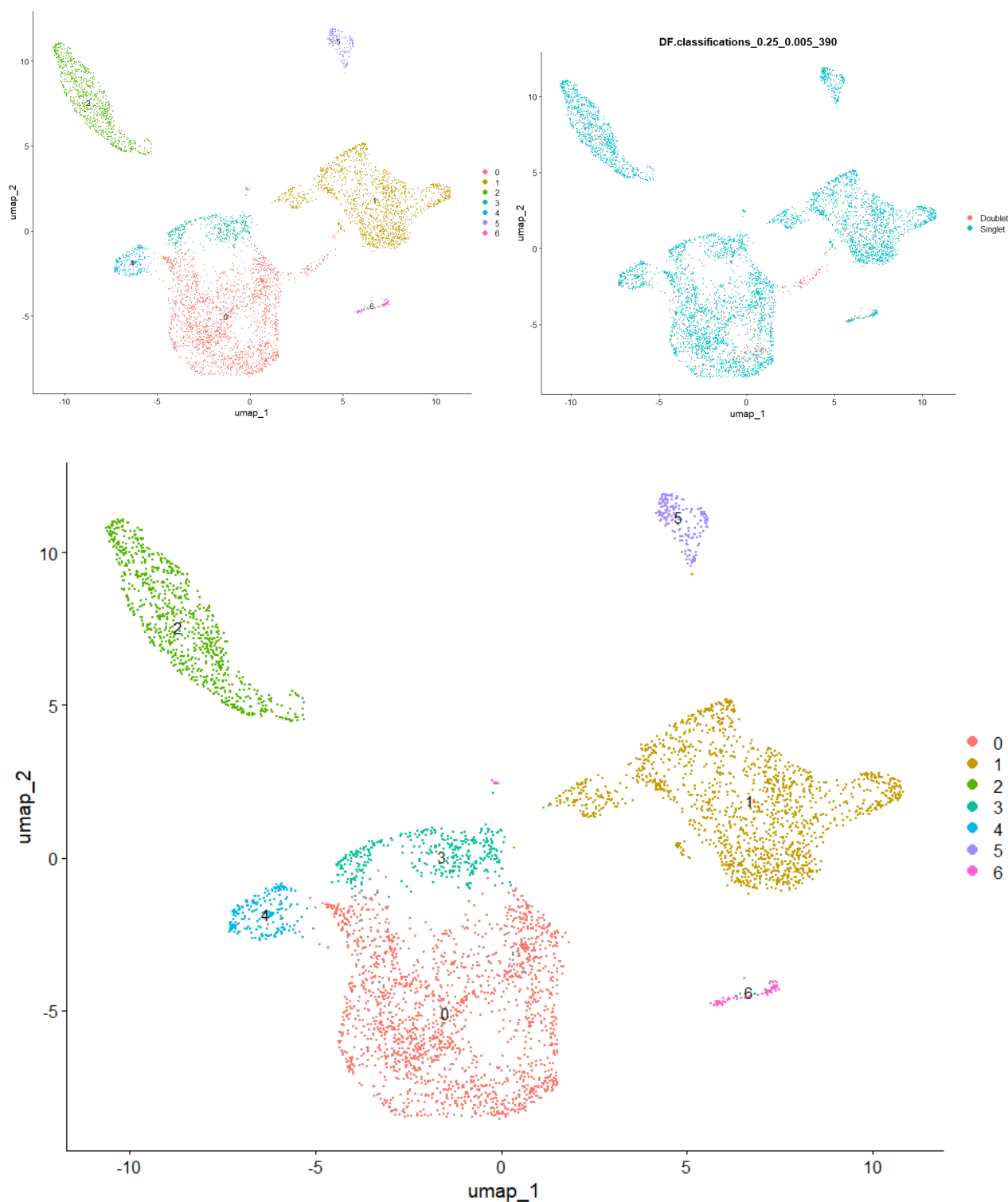


Fig. 2: A – Gene clusters prior to doublet removal. Genes are clustered into seven groups, coded by the key on the right. **B** – Visualisation of likely doublets present within the clusters. Group 0 seems to possess by far the most, with the trailing subgroup to it's right consisting entirely of likely doublets. **C** - Final cluster map with doublets filtered out showing the seven groups.

In Fig. 2 we see the results of clustering the cells based on similarity of differential gene expression. This figure also shows the process of doublet filtering and removal via DoubletFinder. The majority of doublets appeared to exist within cluster zero, and after filtering it remained the most populous cluster. There is small apparent overlap between clusters zero, three and four whilst clusters two, one and five appear well defined. Cluster six is small compared to the others and could perhaps indicate an anomaly, however investigating this would prove difficult without access to more information than is provided on study methodology.

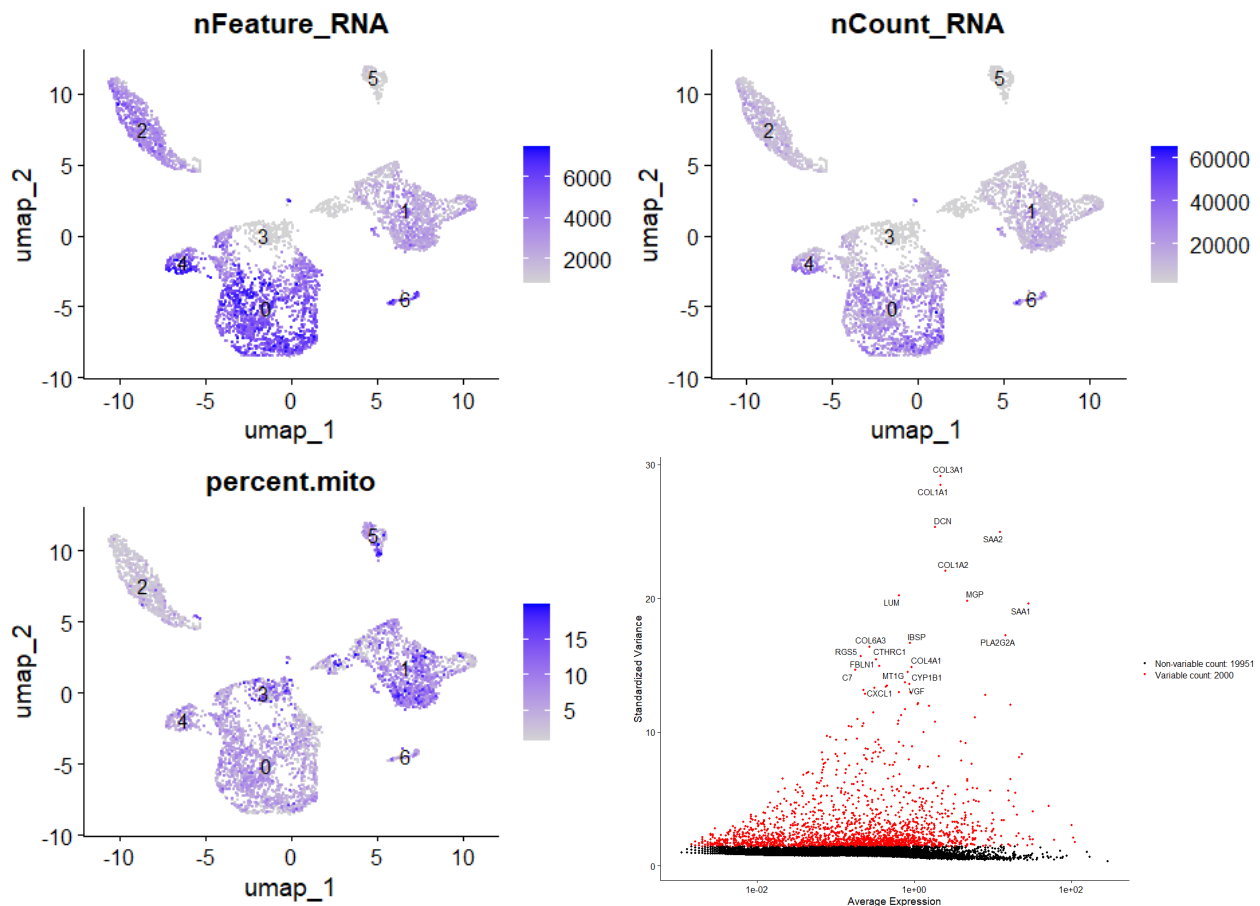


Fig. 3: A – Cluster chart showing the cell clusters alongside the number of features present in each individual cell. **B** - Cluster chart showing the cell clusters alongside the UMIs present in each individual cell. **C** - Cluster chart showing the cell clusters alongside the percentage of mitochondrial RNA present in each individual cell. **D** – Chart showing the top 20 variable features within the dataset after doublets have been removed.

Figures 3A, B and C visualise key metrics for each cluster. Cluster zero has by far the most features per cell on average, whilst count remains relatively even despite clusters three and five showing low numbers. Mitochondrial percentage also remains relatively consistent across populations, however anomalous values will have been filtered out in the preprocessing stage.

Figure 3D shows the top 20 variable genes across the whole cell population after doublet removal. Here these are used for cell annotation and identification, but in a novel study would be worth closer examination. A chart of the top 20 variable genes prior to doublet removal is also included in the figures folder (line 91) within the repository, however the top 20 genes remained unchanged.

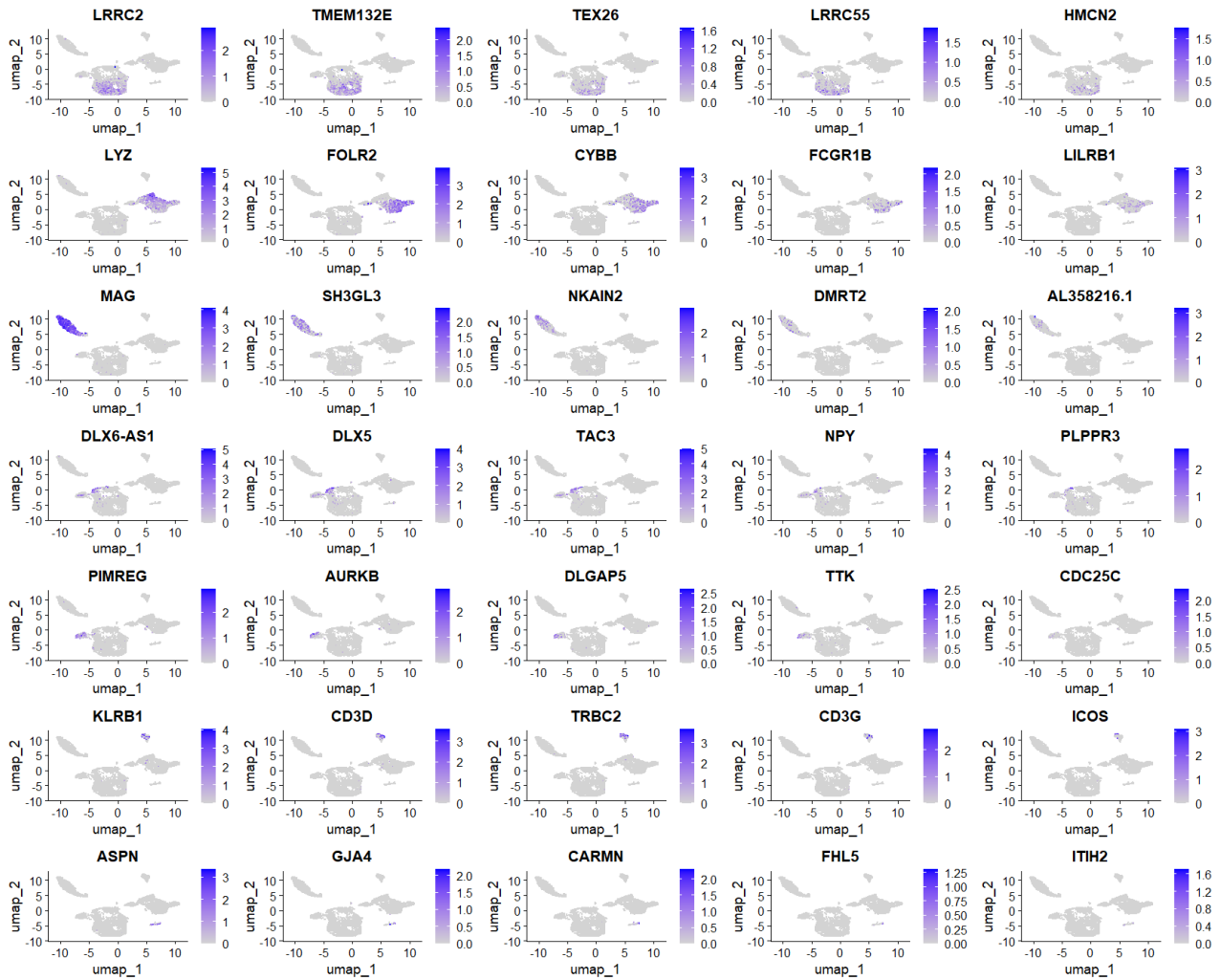


Fig. 4: Charts showing the Top 5 differentially expressed genes per cluster. Each cluster is a row from zero to six, with the level of differential expression determining the column order. Note that this data is also available in the forms of a heatmap and violin plots (lines 235 and 236) as examples depending on preference.

Figure 4 shows some interesting patterns to the differentially expressed genes within the clusters. Within cluster zero, cells towards the bottom of the cluster seem to show the highest levels of the differentially expressed genes, potentially indicating a sub-population of cells within the cluster. This is somewhat reinforced by data seen below in Figure 5.

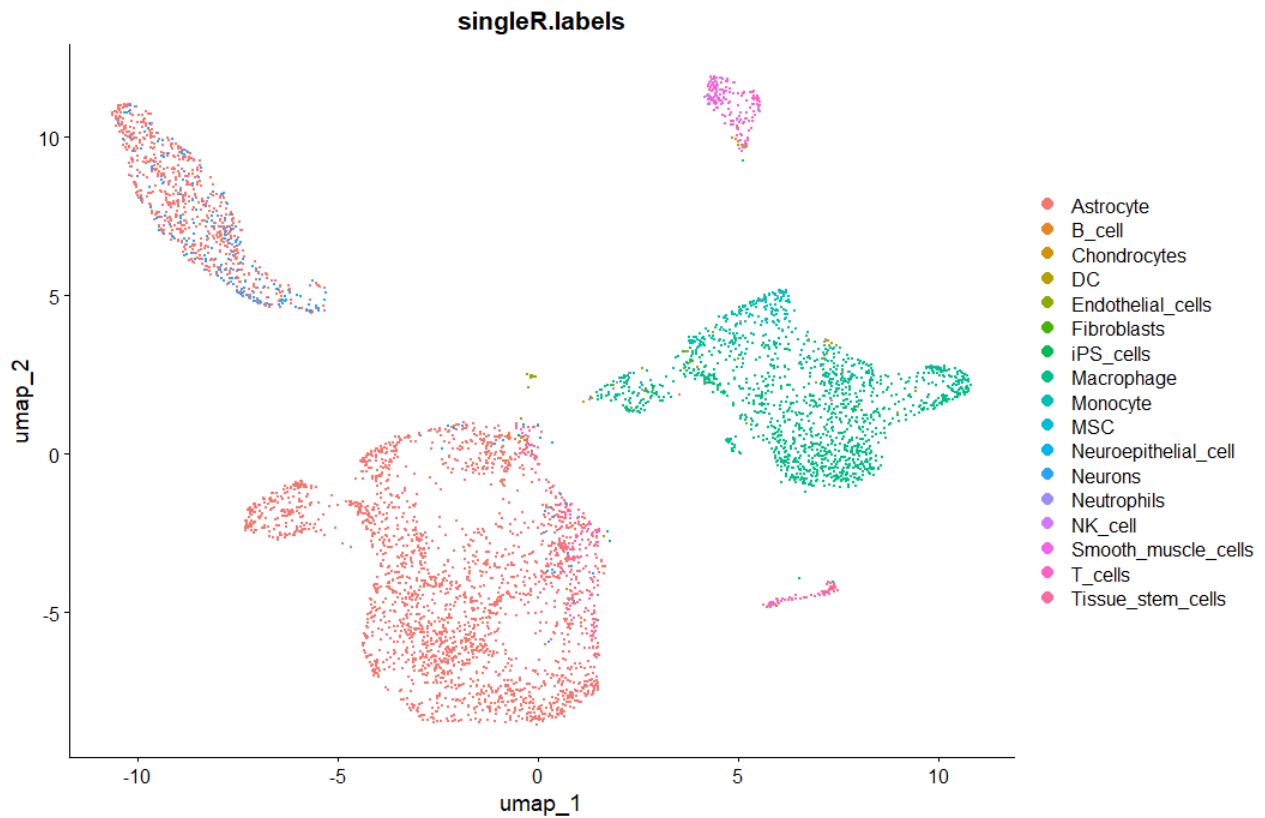


Fig. 5: Chart showing annotated cell types from SingleR alongside clustering. Due to the number of estimated cells present, exact colour matching via hex-codes was used to determine label identities. Also available in the repository figures are some measures of quality for the annotations, however they are not included here as they are not overtly relevant given the data is validated by the published report.

Figure 5 is the best illustration of all of the previous steps, finally showing the clusters annotated with likely cell types using SingleR and CellDex. Cluster zero appears relatively accurate being primarily B Cells with the presence of some astrocytes and few neurones. It also appears that clusters zero, three and four may consist of largely similar cell populations, indicating that the clustering could be relaxed to provide a more representative result. Cluster one appears to largely contain fibroblasts, iPS cells, macrophages, monocytes and MSC cells. Cluster two is somewhat unusual, appearing to contain a mix of astrocytes, B cells and neurons, with a population of B cells displaying different expression patterns to those in cluster zero. Cluster five appears to contain primarily T cells and stem cells, whilst cluster six also appears similar.

DISCUSSION

The results from our data overall align with the results from the published study associated with the dataset, overall demonstrating the validity of the methodology.

Data quality showed an approximately expected number of cells filtered out based on feature, UMI and mitochondrial RNA percentage (7.25%). It was estimated through

DoubletFinder that 390 doublets existed within the data after homotypic adjustment. This leaves our final valid singlet count at 5138 cells.

Cells were distributed into clusters as follows:

CLUSTER NUMBER	CELLS PER CLUSTER
0	1894
1	1550
2	829
3	397
4	202
5	181
6	85

Based on results seen in *Figure 5*, it could be argued that relaxing the clustering threshold to obtain fewer clusters may be more accurate due to the homogeneity seen in clusters zero, four and three. As a counterpoint to this, we also see a mixture of cell types in cluster 2 which may require further definition. The cell labelling we see from SingleR also seems to approximately align with the analysis results of the study, however the study results will be more accurate as a result of manual labelling.

In a typical study, each gene shown in *Figures 3D* and *4* would be thoroughly researched and explained regarding their relationship with the cell population being examined. As this report only aims to outline the methodology used to obtain this data, and the fact that this analysis already exists in the associated publication this lengthy process has been foregone on this occasion.

Overall, this report has described an editable and practical method for obtaining the key results required for ScRNASeq analysis using R, alongside an overview of the various packages required to accomplish this. If seeking to expand on the scope of this analysis, I would encourage examination of the reference material listed on the GitHub readme.